HACKIIITD 2026 • IIIT DELHI

ACCESSPATH AI

“Don’t just test a website. Test whether a real person can actually complete the task.”

Task-Level Accessibility & Usability Testing Powered by a Live AI Browser Agent

THE TASK ACCESSIBILITY JOURNEY WE EVALUATE

1. Diverse User Persona
Mobile, Hindi-speaking, low-literacy

→

2. Real Website
Scholarship or government portal

→

3. End-to-End Task
Multi-step form & document upload

→

4. Success vs. Hidden Barrier
Actionable fix & journey score

Team: [Team Name Placeholder] | Team Members: [Member 1, Member 2, Member 3, Member 4]
HackIIITD 2026 • Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) • TinyFish Track

THE REALITY OF MODERN WEB TESTING

“A Website Can Be Technically Accessible — And Still Impossible to Use.”

WHAT AUDITS REPORT ✅

The Website Appears 100% Ready

- ✓ Passes standard static accessibility checklist
- ✓ HTML contains proper tags and color contrast passes
- ✓ All scholarship application pages load successfully
- ✓ Form input elements exist in DOM structure

HUMAN EXPERIENCE ❌

A First-Time Student Fails to Apply

- ✕ Jargon-heavy instructions confuse Hindi-speaking users
- ✕ Document upload button is hidden inside mobile accordion
- ✕ Silent validation errors leave user stuck without recovery guidance
- ✕ No cue indicates what to do next — application abandoned

The Core Shift: Existing tools ask “Is the website accessible?” ➔ AccessPath asks “Can the user complete the task?”

THE STRATEGIC PARADIGM SHIFT

“Most Tools Check the Website. We Check the Journey.”

TRADITIONAL SCANNERS

Static Page & Code Level

- Evaluates isolated pages: Does not follow end-to-end user journeys
- Static code inspection: Misses dynamic UI state changes and modals
- Generic rule compliance: Scores contrast and tags, not human usability
- Reports technical errors: Developers get 100+ warnings, zero context
- Zero persona awareness: Treats every visitor as an expert desktop user

ACCESSPATH AI

Live Task & Outcome Level

✓ Starts with a real task: “Apply for scholarship”, “Upload ration card”

✓ Live AI browser agent: Executes real clicks, keystrokes, and form steps

✓ Simulates target personas: Mobile, Hindi-speaking, low digital-literacy

✓ Pinpoints exact failure: Identifies step, screen, and cognitive friction

✓ Task Success Score: Actionable, evidence-backed engineering fix

Fundamental Evolution: We move testing from Page-Level Compliance ➔ Task-Level Usability.

PRODUCT ARCHITECTURE

Meet AccessPath AI: The 4-Step Testing Workflow

01

DEFINE

Task & Persona Context

- Target Website URL
- Goal: 'Apply for scholarship'
- Persona: Mobile + Hindi speaker
- Contextual constraints set

02

EXECUTE

Live Agent Run

- TinyFish launches live browser
- Navigates dynamic UI elements
- Fills forms & inputs data
- Attempts task autonomously

03

DETECT

Barrier Recognition

- Flags failed or blocked steps
- Identifies confusing interactions
- Pinpoints language friction
- Catches unrecoverable errors

04

IMPROVE

Actionable Engineering Fix

- Task Success Score (0-100)
- Visual evidence & step logs
- Cognitive barrier explanation
- Recommended UI/UX solution

Live Run Example: Portal URL + 'Apply for Scholarship' + Mobile Hindi Student ➔ Identified mobile upload barrier & score of 62/100.

SYSTEM ARCHITECTURE & EXECUTION PIPELINE

From User Task → Live Website → Actionable Report

LAYER 1 : TEST SPECIFICATION

User & Persona Input

Defines what to test and who is attempting it

- **Website Target**
  Live URL of portal/service
- **User Goal**
  Specific end-to-end task
- **Persona Profile**
  Device, language, digital literacy level

LAYER 2 : LIVE WEB EXECUTION

TinyFish Web Agent Engine

Autonomous live browser interaction on the real web

- **Live Browser Session**
  Direct DOM & rendered UI manipulation
- **Action Stream**
  Click, Type, Select, Scroll, Upload
- **Dynamic Handling**
  Client-side routing, popups, async forms

LAYER 3 : INTELLIGENCE & REPORTING

AccessPath Analysis Engine

Diagnostic reasoning, scoring, and remediation

- **Task Observation**
  Evaluates step progression vs persona friction
- **Barrier Classifier**
  Detects dead ends, visual & language hurdles
- **Scoring & Fix Engine**
  Task Success Score (0-100) + Actionable fix

Architecture in 1 Sentence: TinyFish provides the live web interaction layer; AccessPath provides testing, diagnosis, scoring & remediation.

HACKATHON INTEGRATION FOCUS

Why TinyFish Is Essential: Clear Division of Responsibility

TINYFISH AGENT LAYER

“Eyes & Hands on the Web”

Live Execution & Real Browser Traversal

- Live website navigation across multi-page flows
- Multi-step interactive browser operations
- Clicking, typing, dropdown selection & uploads
- Handling dynamic client-rendered applications (SPAs)
- Executing tasks exactly as a human agent would

ACCESSPATH INTELLIGENCE LAYER

“Testing, Diagnosis & Decision Layer”

Cognitive Evaluation, Scoring & Insights

- Task & user persona definition parameters
- Cognitive barrier detection (dead ends, confusion)
- Quantitative Task Success Score calculation (0–100)
- Evidence capture: screenshot, step traces & log context
- Actionable remediation & UI fix generation

The Synergy: TinyFish executes the task in real-time ➔ AccessPath evaluates whether the user succeeded, why they failed, and how to fix it.

WORKING PROTOTYPE & HACKATHON DEMO

Our 24-Hour MVP: Feasible, Focused, and Functional

MVP TEST RUN INPUT

Target URL: scholarship.state.gov.in (Portal)

Goal Task: “Complete Scholarship Application”

Persona: Mobile viewport + Hindi preference

24-HOUR AUTOMATED PIPELINE

1. Launch TinyFish agent with mobile viewport
2. Attempt multi-step registration workflow
3. Track DOM interaction & UI response times
4. Intercept barrier at Step 9 (Document upload)
5. Compile audit payload & calculate score

ACCESSPATH AUDIT CONSOLE • TEST RUN ID: #AP-8842

TASK SUCCESS SCORE

62 / 100

STATUS: BLOCKED

STEP BREAKDOWN

✅ Completed Steps: 8 / 11 steps successful (Landing, Form fields)

⚠️ Blocked Friction: 2 interactions with significant delay

🔴 Critical Barrier: 1 unrecoverable blocker at upload step

IDENTIFIED BARRIER & RECOMMENDED FIX

🔴 **Critical Barrier: Document upload control collapsed inside hidden mobile accordion.**

Evidence: On 375px mobile viewport, upload input lacks visible label in Hindi; user is offered no feedback when file format rejects silently.

💡 **Recommended Fix: Un-collapse upload control by default on mobile viewports. Provide explicit bilingual format hints (PDF/JPEG < 2MB) and prominent error toast.**

MARKET DIFFERENTIATION & VIABILITY

Why AccessPath Can Matter: Value, Reach, and Roadmap

UNIQUE ADVANTAGES

Core Differentiators

✓ Task-First Testing
Tests what users actually need to accomplish, not isolated syntax rules.

✓ Persona-Aware
Recognizes that mobile, elderly, and rural users experience the same site differently.

✓ Live-Web Autonomous Agent
Executes multi-step browser workflows via TinyFish, capturing dynamic state friction.

✓ Actionable Remediation
Reports exact step failure with evidence and engineering-ready fixes.

TARGET SECTORS

High-Stakes Portals

🏛 Government Public Portals
Scholarships, ration cards, licenses, and DBT welfare distribution.

🏛 Universities & Colleges
Admissions, entrance test applications, and fee payment portals.

🏛 Banking & FinTech
Digital KYC onboarding, micro-credit applications, insurance claims.

🏛 Healthcare & Telemedicine
Appointment booking, diagnostic report downloads, health IDs.

GROWTH ROADMAP

Phased Scalability

➔ Phase 1: 24-Hour Prototype
Single task, single website target, automated barrier identification.

➔ Phase 2: Multi-Persona Matrix
Batch evaluation across mobile, regional languages & assistive profiles.

➔ Phase 3: CI/CD Pipeline Plug-in
Automated accessibility regression testing on pull requests.

➔ Phase 4: Continuous Enterprise
Organization-wide accessibility health monitoring across full web estate.

VISION & MISSION

“Accessibility Should Be Measured by Outcomes.”

TRADITIONAL CHECKLIST TESTING

“Does the website pass static technical rules?”

→

ACCESSPATH AI OUTCOME TESTING

“Can a real person actually complete the task?”

WHAT ORGANIZATIONS PROVIDE

Live Website URL + Target Task + User Persona Profile

WHAT ACCESSPATH DELIVERS

Task Success Score + Step Evidence + Actionable UI Fix

“AccessPath AI helps organizations build websites that people can actually use — not just websites that pass a checklist.”

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